

WEEK 7 PROJECT SUMMARY — HEALTHCONNECT (DATA ANALYTICS TRACK)

1. **Planned to test:** the accuracy of my top KPIs (Python vs a manual Excel check), my dashboard charts and filters for misleading axes or broken behaviour, and whether the Data Science track's model definitions still match mine.
2. **Actually tested:** KPI 1 (no-show rate), KPI 3 (repeat-misser concentration), Chart 1 (lead time $\times$ reminder), the distance filter behind the telehealth chart, and the cross-track target definition using the Data Science Week 6 questionnaire.
3. **Most important testing results:** the KPI math matched Excel exactly (48.5% overall; 51.1% excluding cancellations); the repeat-misser ratio held steady on a random 20% sample; the cross-track definition check passed, both tracks now exclude cancellations.
4. **Issues or weaknesses identified:** Chart 1's Y-axis plotted raw lead days instead of percentages and used one default colour, making it misleading; the 30+ day reminder subgroup is small, so its 4-point gap is directionally right but not precise.
5. **Improvements/refinements made:** rewrote the Chart 1 code to plot percentages with a green/red palette; added subgroup sizes next to every interaction percentage in the report; reworded the recommendations to say "pilot before rollout."
6. **Retesting results:** Chart 1 passed on retest — title, axis and colours now agree; the KPI and filter tests passed first time and were re-verified after the chart fix.
7. **Track collaborated with:** Data Science.
8. **What was tested collaboratively:** the end-to-end definition alignment — my KPI 1 formula against their binary target and their feature list (`is_long_distance`, `no_show_rate`).
9. **What changed as a result:** confirmation that both tracks measure no-shows identically, and my recommendation now uses their model's probability score to auto-flag high-risk patients for the phone-call intervention.
10. **Key findings or validation outcomes:** the three drivers (lead time, history, distance) are mathematically verified and stable across samples; reminders remain a modest lever; the analytics component is ready for Week 8 integration.
11. **Major challenges:** reproducing the Python cuts by hand in Excel was slow; deciding how to report small interaction subgroups without overstating them.
12. **Important decisions:** keep every test reproducible (the Excel pivot steps are documented in the notebook); report subgroup sizes beside percentages; document successful validations honestly instead of inventing failures.
13. **Remaining limitations:** synthetic data with an unrealistically high no-show rate; 90 missing distance and 60 missing waiting-time values; no causal evidence.

14. **Remaining issues or dependencies:** Data Science's Week 6 model run is still pending, so feature-importance confirmation and the probability-score handover must wait; the telehealth recommendation assumes the clinic can offer video consultations.
15. **Contribution to the overall HealthConnect project:** a tested, verified evidence base — validated KPIs, confirmed drivers and corrected charts — that the model, the dashboard and the final presentation can all rely on safely.
16. **Must be completed before Week 8:** combine my validated business rules with Data Science's final model into one intervention strategy; build the final dashboard and presentation slides; close the pending feature-importance check.